Exchange Rate Regimes in Production Networks

A Small Open Economy Extension of Rubbo (2024)

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Building on: Rubbo (2024) *Networks, Phillips Curves, and Monetary Policy*

Motivation: Networks Meet the Exchange Rate

Two literatures that rarely talk to each other:

1. **Production-network Phillips curves** (Rubbo 2024; La'O & Tahbaz-Salehi 2022): with input-output linkages, sector-level price stickiness generates a genuine inflation–output tradeoff for *almost every* price index — except one. A unique "divine coincidence" (DC) index, weighting sectors by Domar-scaled stickiness, restores the tradeoff-free policy prescription. All of this is derived in a **closed economy**.
2. **Small open economy NK models** (Galí & Monacelli 2005 and successors): the exchange rate is a cost-push shifter and a policy instrument, but the production side is a single aggregate sector — no network structure, no heterogeneity in who is exposed to FX shocks.

This paper puts them together. Once imported inputs enter a multi-sector network, the exchange rate becomes a *second* cost-push channel that hits sectors asymmetrically depending on their import centrality and their position in the domestic supply chain. Does the DC index — or any analog — survive?

Why This Matters: The Policy Question

Motivating case: a small oil-exporting economy. Resource sector is upstream, import-intensive (machinery) and export-heavy; Manufacturing sits in the middle; Services is downstream, domestically oriented, and dominates final consumption.

The question a central bank actually faces:

- Float and target inflation $\Rightarrow$ exchange rate absorbs FX/commodity-price shocks, but at the cost of nominal volatility and possibly balance-sheet risk.
- Peg $\Rightarrow$ eliminates FX risk, but forces *all* adjustment onto domestic prices and output — amplified by whichever sectors are most import-exposed and most central in the network.
- Manage the float $\Rightarrow$ partially, but at what welfare cost relative to the theoretical optimum?

Rubbo's closed-economy answer (target the DC index) says nothing about *which* exchange-rate regime to combine it with, or how network position shapes the answer. That is the gap this paper fills.

Literature

- **La'O & Tahbaz-Salehi (2022, *Econometrica*)** — optimal monetary policy in production networks; efficient allocations require targeting a Domar-weighted index, not simple CPI. Closed economy.
- **Rubbo (2024, *Econometrica*)** — the paper we extend. Sectoral Calvo pricing + IO network $\Rightarrow$ unique DC index with *no* cost-push residual; welfare-optimal Taylor rule targets it.
- **Auer, Levchenko & Sauré (2019, *ReStud*)** — international inflation spillovers propagate through global IO linkages; empirical evidence that network position matters for cross-border pass-through.
- **Pasten, Schoenle & Weber (2020, *JPE*)** — heterogeneous price rigidity across sectors reshapes monetary transmission; motivates letting $\hat{\delta}_i$ differ by sector here too.
- **Fanelli & Straub (2021, *JPE*)** — theory of sterilized FX intervention as a second instrument alongside the policy rate.
- **Gopinath & Itskhoki (2022)** — dominant currency pricing; **Amiti, Itskhoki & Konings (2019, *QJE*)** — variable markups shape exchange-rate pass-through at the firm level.

Gap: no existing paper combines a multi-sector Calvo network *and* a genuine open-economy block (UIP, imported inputs, FX regimes) to ask whether the DC-targeting result survives.

Rubbo (2024): What Carries Over

Closed-economy system (N sectors, Calvo $\hat{\delta}_i$, IO network Ω):

$$
\begin{aligned}
\text{IS:} \quad & \tilde{y}_t = \mathbb{E}\tilde{y}_{t+1} - \tfrac{1}{\gamma}(i_t - \mathbb{E}\pi^C_{t+1} - r^{nat}_t) \\
\text{MC:} \quad & \boldsymbol{mc}_t = \boldsymbol{\alpha} w_t + \Omega \boldsymbol{p}_t - \log \mathbf{A}_t \\
\text{NKPC:} \quad & \boldsymbol{\pi}_t = \rho(I - \mathcal{V})\mathbb{E}\boldsymbol{\pi}_{t+1} + \mathcal{B}(\gamma+\varphi)\tilde{y}_t - \mathcal{V}\boldsymbol{\chi}_t \\
\text{Policy:} \quad & i_t = r^{nat}_t + \phi_\pi \pi^C_t + \phi_y \tilde{y}_t
\end{aligned}
$$

Lemma 1: natural output = Domar-weighted TFP, $y^{nat}_t = \sum_i \lambda_i a_{it}$.

Proposition 1 (DC index): unique index with *no* cost-push:

$$\pi^{DC}_t = \rho\, \mathbb{E}\pi^{DC}_{t+1} + \tfrac{\gamma+\varphi}{\Lambda^{DC}}\tilde{y}_t, \quad w^{DC}_i \propto \lambda_i \tfrac{1-\hat{\delta}_i}{\hat{\delta}_i}$$

Why DC is unique: $\mathcal{V} = \Delta(I - \Omega\Delta)^{-1}$ has rank $N{-}1$ on the space of cost-push shocks $\Rightarrow$ a one-dimensional null space $\Rightarrow$ exactly one index ϕ with $\phi^\top \mathcal{V} = \mathbf{0}^\top$.

One shock, one channel: in the closed economy, only TFP shocks $\boldsymbol{\chi}_t$ enter the cost-push term. That is exactly why a single index can null it out.

The open question we ask: with a *second* cost-push channel (the exchange rate), can the same trick still null out both simultaneously?

This Paper: Contribution and Approach

What we do:

- Extend Rubbo's network Phillips curve to a small open economy: imported intermediate inputs (with an import-cost-share matrix Ω^F), imported final consumption, UIP with a debt-elastic premium.
- Ask whether the DC index survives when the exchange rate is a second cost-push channel, and quantify the welfare cost of float / peg / managed-float regimes.
- Calibrate to a 3-sector oil-exporting economy (Resource $\rightarrow$ Manufacturing $\rightarrow$ Services).

Methodological choice: the model is built and solved as the *original nonlinear* system — Cobb-Douglas cost minimization, the exact recursive Calvo pricing problem, CRRA/Frisch household FOCs, CES consumption aggregation — rather than substituting in a hand-linearized Phillips curve. The nonlinear steady state is computed numerically and the system perturbed around it, so the log-linear coefficients (e.g. $\hat{\delta}_i$) come out of the solution rather than being assumed.

Model Setup: Households

Preferences (CRRA consumption, Frisch labor supply; King, Plosser & Rebelo 1988):

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\gamma}}{1-\gamma} - \frac{L_t^{1+\varphi}}{1+\varphi} \right]$$

Consumption aggregation: CES nest between a domestic bundle and an imported bundle, home-bias weight ω, elasticity η (Armington 1969; Galí & Monacelli 2005); the domestic bundle is itself Cobb-Douglas across the N sectors with shares β_i^H:

$$C_t = \left[\omega^{1/\eta} (C_t^H)^{\frac{\eta-1}{\eta}} + (1-\omega)^{1/\eta} (C_t^F)^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}}, \qquad C_t^H = \prod_i \left(\frac{C_{it}}{\beta_i^H} \right)^{\beta_i^H}$$

Household FOCs (exact nonlinear conditions):

$$\text{Euler:} \quad C_t^{-\gamma} = \beta(1+i_t)\,\mathbb{E}_t\left[C_{t+1}^{-\gamma} / \Pi_{t+1}^C \right]$$

$$\text{Labor supply:} \quad W_t/P_t^C = C_t^{\gamma} L_t^{\varphi}$$

Asset side: single internationally traded bond B_t^*; no domestic bond needed (representative household, net-zero supply, redundant).

Model Setup: Firms and Calvo Pricing

Production (Cobb-Douglas, sector i; network structure as in La'O & Tahbaz-Salehi 2022, Rubbo 2024):

$$Y_{it} = A_{it}\, L_{it}^{\alpha_i} \prod_j X_{ijt}^{\Omega_{ij}^H}\, M_{it}^{\Omega_i^F}$$

Cost minimization $\Rightarrow$ nominal marginal cost is multiplicatively separable:

$$MC_{it} = \frac{W_t^{\alpha_i} \prod_j P_{jt}^{\Omega_{ij}^H}\, (S_t P_t^{F*})^{\Omega_i^F}}{A_{it}}$$

Calvo pricing (Calvo 1983) — fraction δ_i resets each period (*not* $\hat{\delta}_i$: that is derived, not primitive):

$$P_{it}^{\#} = \frac{\varepsilon}{\varepsilon - 1} \cdot \frac{\mathbb{E}_t \sum_s \rho^s (1-\delta_i)^s Y_{it+s}\, \widetilde{MC}_{it+s}}{\mathbb{E}_t \sum_s \rho^s (1-\delta_i)^s Y_{it+s}/P_{it+s}}$$

Exact nonlinear recursion (real MC $\widetilde{MC} = MC/P$):

$$\begin{aligned}
X1_{it} &= \widetilde{MC}_{it} Y_{it} + (1-\delta_i)\beta \left(\tfrac{C_{t+1}}{C_t}\right)^{-\gamma} \Pi_{i,t+1}^{\varepsilon} X1_{i,t+1} \\
X2_{it} &= Y_{it} + (1-\delta_i)\beta \left(\tfrac{C_{t+1}}{C_t}\right)^{-\gamma} \Pi_{i,t+1}^{\varepsilon-1} X2_{i,t+1} \\
P_{it}^{\#}/P_{it} &= \tfrac{\varepsilon}{\varepsilon-1}\, X1_{it}/X2_{it} \\
1 &= (1-\delta_i)\Pi_{it}^{\varepsilon-1} + \delta_i (P_{it}^{\#}/P_{it})^{1-\varepsilon}
\end{aligned}$$

Why this matters: log-linearizing this exact recursion is what *produces* Rubbo's adjusted parameter $\hat{\delta}_i = \frac{\delta_i(1-\beta(1-\delta_i))}{1-\beta\delta_i(1-\delta_i)}$ — it is a derived object, not a primitive of the model.

Model Setup: The Open-Economy Block

Exchange rate and imports (S_t: nominal ER, P_t^{F*}: foreign import price, law of one price; Galí & Monacelli 2005):

$$P_t^{FH} = S_t P_t^{F*}$$

UIP with a debt-elastic premium (Schmitt-Grohé & Uribe 2003, for stationarity):

$$(1 + i_t) = (1 + i^*)\left[1 - \psi(\hat{b}_t^* - \bar{b}^*)\right] \mathbb{E}_t \left[S_{t+1}/S_t\right]$$

Net foreign assets (current-account identity):

$$\hat{b}_t^* = \frac{1 + i_{t-1}}{\Pi_t^C} \hat{b}_{t-1}^* + \frac{P_t^H EX_t - P_t^{FH} IM_t}{P_t^C}$$

Trade flows:

$$EX_t = \kappa_{EX} D_t^* \left(\frac{P_t^H}{P_t^{FH}}\right)^{-\theta^*}, \qquad IM_t = C_t^F + \sum_i \Omega_i^F \frac{MC_{it} Y_{it}}{P_t^{FH}}$$

Key mechanism: imported inputs enter every sector's marginal cost through Ω_i^F — the exchange rate now moves marginal cost *exactly like a productivity shock*, scaled by each sector's import exposure and propagated through Ω^H.

Policy Regimes and Shocks

Three FX regimes (only the policy block changes):

Regime	**Rule**	**Exchange rate**
Float	$i_t = \bar{\imath} + \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t$	fully endogenous (UIP)
Peg	$S_t = \bar{S}$	fixed; i_t residual from UIP
Managed float	$i_t = \bar{\imath} + \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t + \phi_s \log(S_t/\bar{S})$	partially stabilized

Three shocks studied:

- Sectoral TFP shocks a_{it}, AR(1), $\rho_a = 0.90$
- Import price shock p_t^{F*}, AR(1), $\rho_{pF} = 0.85$ — the pure FX/commodity-price channel
- Foreign demand shock D_t^*, AR(1), $\rho_D = 0.80$ — affects exports, hence NFA and the exchange rate

Solution Method: Nonlinear, Not Pre-Linearized

Why solve the nonlinear model directly, rather than hand-linearize first:

- The Calvo slope $\hat{\delta}_i$ and the DC weights are *outputs* of log-linearizing the true recursion — coding them as inputs risks baking in the answer.
- A genuine nonlinear model lets us go to order-2 perturbation, which is required for valid welfare comparisons across policy regimes (Kim-Kim / Woodford: first-order-accurate welfare rankings are not valid — variance terms enter the welfare function only at second order).

Steady state: solved semi-analytically, not guessed. At zero inflation, price = markup × marginal cost exactly, which collapses the 46-equation system to one scalar root-find in the real wage W (prices, exports, consumption all closed-form given W), giving $W^* = 1.3189$.

- **Lemma 1 (exact):** $\tilde{y}_t = \sum_i \lambda_{D,i} \left[\log(Y_{it}/\bar{Y}_i) - \log A_{it} \right]$ — no shadow flex-price block needed.
- **Prop. 1 weights (theory-motivated):** DC weights from $\lambda_{D,i}$, $\hat{\delta}_i$, applied to realized inflation in the true nonlinear simulation.

Solved at first order around the exact steady state, for all three regimes. Second-order perturbation is not used: an inflation/output-gap-only Taylor rule leaves the price *level* and S_t as a unit root — expected, since nothing anchors the level — so only the first-order (stationary) solution is used, and only stationary objects enter welfare.

Calibration: A 3-Sector Oil-Exporting Economy

Parameter	Value	Parameter	Value
β (discount factor)	0.99	Ω^H_{21} (Manuf. $\leftarrow$ Resource)	0.20
γ (risk aversion)	1.00	Ω^H_{32} (Services $\leftarrow$ Manuf.)	0.25
φ (inv. Frisch elasticity)	2.00	$\Omega^F_1, \Omega^F_2, \Omega^F_3$ (import shares)	0.30, 0.10, 0.05
η (home/foreign elasticity)	1.50	$\beta^H_1, \beta^H_2, \beta^H_3$ (consumption shares)	0.05, 0.15, 0.80
$\beta^{F,tot}$ (import share, CPI)	0.10	$\delta_1, \delta_2, \delta_3$ (Calvo reset prob.)	0.75, 0.50, 0.25
ε (within-sector elasticity)	8.00	ψ (NFA debt-elastic premium)	0.020
θ^* (export price elasticity)	2.00	ϕ_π, ϕ_y, ϕ_s (policy)	1.50, 0.50, 0.30

Sector story: Resource is upstream, flexible-priced, import-intensive (machinery) and the main exporter. Manufacturing is the middle of the chain. Services is downstream, sticky-priced, and dominates final consumption — the classic setting for network amplification.

Network Properties: Centrality and DC Weights

Domestic supplier centrality (Domar weight):

$$\lambda_{D,i} = \left[(\beta^H)^\top (I - \Omega^H)^{-1} \right]_i$$

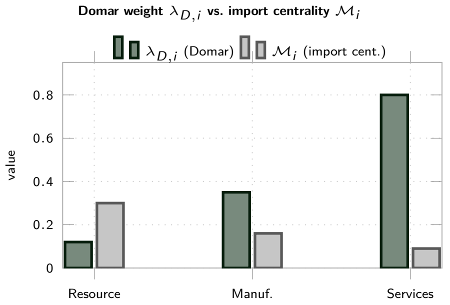
Import centrality (FX exposure transmitted through the domestic network):

$$\mathcal{M}_i = \left[(I - \Omega^H)^{-1} \Omega^F \mathbf{1} \right]_i$$

DC-index weights (Proposition 1, applied here to the true nonlinear realized inflation):

$$w_i^{DC} \propto \lambda_{D,i} \frac{1 - \hat{\delta}_i}{\hat{\delta}_i}$$

Both objects are computed once from the calibrated $\Omega^H, \Omega^F, \beta^H$ matrices — no magic numbers, same formulas as Rubbo's Proposition 1 and the import-centrality analog.



Expect: Services has the largest $\lambda_{D,i}$ (dominates final demand, downstream); Resource has the largest $\mathcal{M}_i$ (directly import-intensive) but this need not be the sector with the largest DC weight, since DC weights also load on stickiness $(1 - \hat{\delta}_i)/\hat{\delta}_i$.

Open-Economy Divine Coincidence: Does It Survive?

Closed economy (Rubbo): one cost-push channel (TFP) $\Rightarrow$ DC index exists uniquely because a rank-deficient matrix $\mathcal{V}$ has a one-dimensional null space, and one index can sit in it.

Open economy: two cost-push channels. TFP *and* the exchange rate now shift sectoral marginal cost. A single weight vector would need to null out *both* simultaneously:

$$\underbrace{\phi^\top \mathcal{V} = \mathbf{0}^\top}_{\text{TFP channel}} \quad \text{and} \quad \underbrace{\phi^\top \Gamma = 0}_{\text{FX channel}}$$

Two conditions, one free vector $\Rightarrow$ generically no solution: **DC breaks down under a float.**

But: import nodes can be modeled as fully flexible upstream sectors ($\hat{\delta}_M = 1$) in an augmented network. Fully flexible nodes get *zero* DC weight ($w_M = 0$ since $(1 - \hat{\delta}_M)/\hat{\delta}_M = 0$), so Rubbo's Proposition 1 still applies to the *augmented* network — the DC index (as coded here) remains well-defined and insulates against domestic TFP shocks by construction. Whether it *also* nulls the FX channel in the calibrated network is the quantitative question tackled below.

Regime Comparison: Theoretical Predictions

	Float	**Peg**	**Managed**
Exchange rate	fully endogenous	fixed	partially stabilized
FX cost-push absorbed by	S_t adjustment	domestic π_i, y	split
Monetary independence	full	none (residual i_t)	full
Pass-through of import shocks	low (if ϕ_π high)	proportional to $\mathcal{M}_i$	intermediate
DC index well defined?	yes (augmented network)	n/a (S_t not a choice)	yes

Naive expectation: float should dominate on inflation stabilization since it directly targets π_t^{DC}; peg transmits import shocks directly, scaled by $\mathcal{M}_i$; managed sits in between.

Preview: the quantitative results overturn part of this. Solving the actual nonlinear model shows free float is *not* the best regime here — see Results.

From Theory to Quantification

Pipeline: the model is solved numerically for all three regimes, producing IRFs and theoretical moments, summarized in the figures below. Every number on the following slides is model output for the calibration two slides ago.

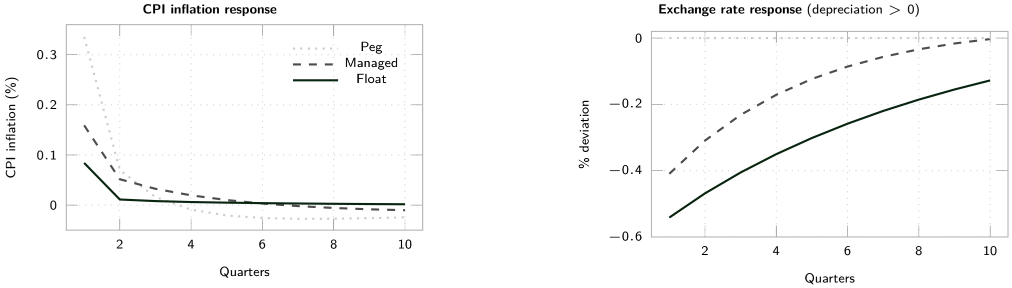
Six questions the simulations answer:

1. How do sectoral inflation, the output gap, and the exchange rate respond to TFP, import-price, and foreign-demand shocks, under each regime?
2. Which sectors are most exposed, and does that line up with Domar weight, import centrality, or both?
3. How do exports and imports move, and what does that imply for the trade balance and NFA dynamics?
4. Does the DC index stay close to zero across shocks and regimes (insulation property), or does it visibly move under the float?
5. What is the welfare ranking of float vs. peg vs. managed float?
6. How much of the welfare cost is output-gap volatility vs. sector-level price dispersion?

Headline (surprising) answer: free float is *not* the best regime here. Managed float dominates on almost every margin, and even the hard peg beats pure float on *total* welfare. The next slides show why.

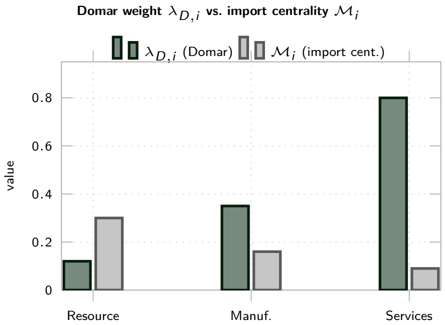
Results: IRFs to an Import-Price Shock

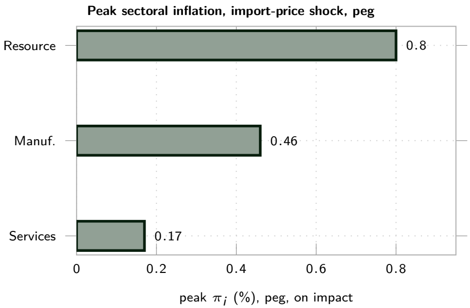
A 1% increase in the foreign-currency import price p_t^{F*} (persistence $\rho_{pF} = 0.85$).



Reading: the currency *appreciates* (S_t falls) to fight the imported cost-push shock — not the naive "let it depreciate and absorb the shock" story. Peg has by far the largest on-impact CPI response (0.34%) but it *fully reverses* within 3–4 quarters (overshoot-correction); float's response is small on impact (0.08%) but decays far more slowly. Persistence, not just amplitude, is what ends up mattering for welfare (next slides).

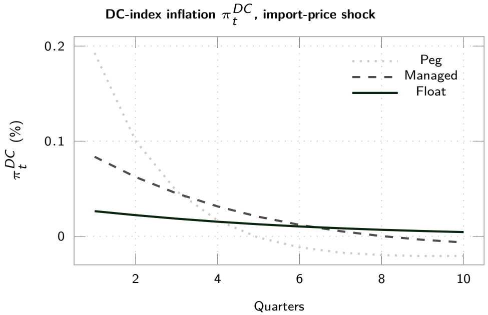
Results: Sectoral Heterogeneity and Network Centrality





Two different rankings, confirmed by simulation: Resource has the highest import centrality $\mathcal{M}_i = 0.30$ and (as predicted) the most volatile own inflation on impact (0.80%), but the lowest Domar weight $\lambda_{D,i} = 0.12$ — its shocks barely move the aggregate. Services is the mirror image: low import exposure ($\mathcal{M}_3 = 0.09$), the least volatile own inflation on impact (0.17%), but dominates the aggregate (Domar weight 0.80, DC weight 0.93). As the welfare numbers below show, "least volatile on impact" does *not* mean "least costly" once persistence is priced in.

Results: DC-Index Insulation and Trade Flows



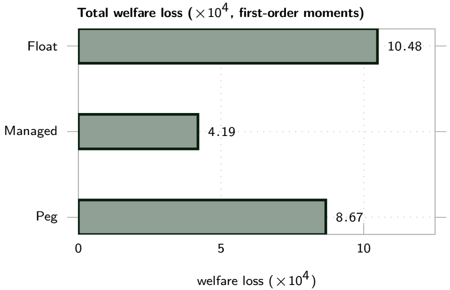
Peak response to a foreign-demand shock

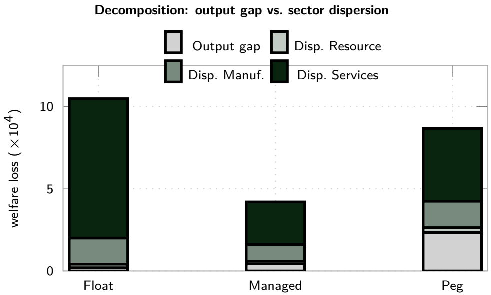
(%):		Float	Managed	Peg
	Exports EX_t	0.122	0.134	0.182
	Imports IM_t	0.081	0.090	0.130
	NFA $\hat{b}_t^*$	0.122	0.122	0.119
	Output gap	0.064	0.115	0.339

NFA accumulation is almost identical across regimes; the output gap is not — peg's output gap moves $\sim 5\times$ more than float's for the same demand shock, since the interest rate cannot help absorb it.

DC insulation, on impact: float's π_t^{DC} is smallest on impact (0.026%) — the augmented-network DC index does dampen the immediate FX cost-push. But float's response is also the *most persistent* (barely decaying by quarter 10), while peg's and managed's both overshoot and reverse sign. On-impact insulation is not the whole story.

Results: Welfare Cost by Regime — The Headline Finding

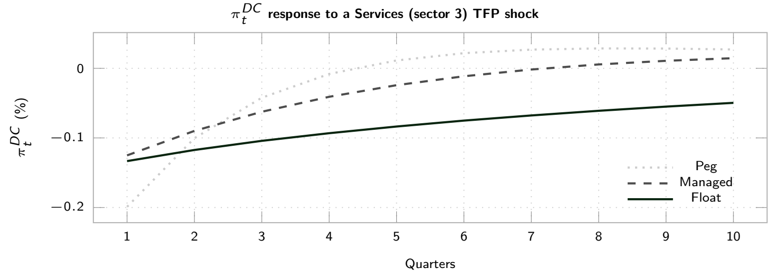




Welfare measure (Rubbo Prop. 3, adapted): $\mathcal{W} = \frac{1}{2}\left[(\gamma + \varphi)\mathrm{Var}(\tilde{y}_t) + \sum_i \lambda_{D,i}\varepsilon_i \frac{1-\hat{\delta}_i}{\hat{\delta}_i}\mathrm{Var}(\pi_{it})\right]$, evaluated on Dynare's own order-1 theoretical variances (see caveat on order vs. welfare, two slides back). **Float has the lowest output-gap cost of all three (0.19) but by far the largest Services-dispersion cost (8.48, driven by the persistence channel above) — managed float is the only regime that is not dominated on some margin.**

Why Floating Costs More Than Expected: Persistence, Not Volatility

Mechanism: under float, S_t is pinned down by UIP and the (highly persistent, autocorrelation 0.994) net-foreign-asset process $\hat{b}_t^*$. Since S_t enters *every* sector's marginal cost via Ω_i^F, letting the currency float doesn't just insulate against FX shocks — it also lets *domestic* TFP shocks propagate through a new, slow-decaying exchange-rate channel that does not exist under a peg ($S_t \equiv 0$).



Reading: peg resolves the shock via a fast, one-time adjustment and overshoots/reverses within 3–4 quarters; float's response barely decays over the whole horizon. This is a *pure domestic TFP shock* — no FX shock to insulate against — yet float still has the most drawn-out response, since the exchange rate is now a slow state variable feeding continuously back into costs.

Which Sector Prefers Which Regime? Two Equations

Two calibratable objects, not one, determine a sector's regime preference — and in this economy they point in *opposite* directions across sectors.

1. Direct FX exposure (network- and rigidity-weighted import share):

$$\Gamma_i = \frac{[\Delta(I - \Omega\Delta)^{-1}\omega_F]_i}{\kappa}$$

2. Welfare weight (centrality × stickiness):

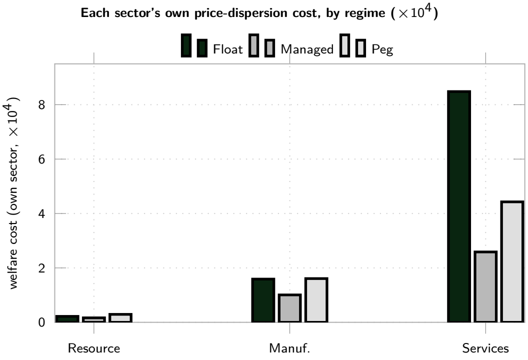
$$\kappa_i = \lambda_{D,i}\,\varepsilon\,\frac{1-\hat{\delta}_i}{\hat{\delta}_i}$$

	Resource	Manuf.	Services
Γ_i (FX exposure)	**0.234**	0.054	0.006
κ_i (welfare weight)	0.43	5.54	**74.6**

The sector most exposed to the exchange rate on impact is the one welfare cares about least; the sector welfare cares about most is almost insulated from it directly. A peg switches off $\Gamma_i \Delta e_t$ for everyone — but it does not eliminate adjustment, it *relocates* it to domestic prices, landing on the upstream sector because it sits at the border (P^H is dominated by its price).

Net effect (real $\kappa_i \mathrm{Var}^R(\pi_i)$, $\times 10^4$): Resource's relocated-adjustment cost under peg outweighs its avoided Γ-cost (0.44 → 0.59) — so despite the largest Γ_i, it still prefers *float*. Services is insulated from both channels, so peg nearly halves its inflation variance (16.96 → 8.86) — and since $\kappa_3 \gg \kappa_1, \kappa_2$, this one sector's preference dominates the aggregate ranking.

Sector-Level Winners and Losers



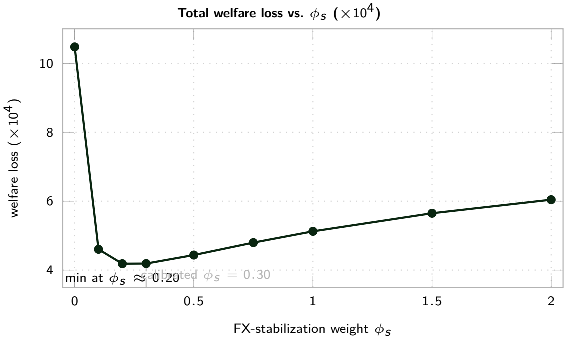
All three sectors agree on their first choice: managed float. They disagree on the runner-up:

- **Resource** (flexible, peripheral, most import-exposed): Managed > Float > Peg. Direct FX exposure makes the peg its worst option.
- **Manufacturing**: Managed $\gg$ {Float $\approx$ Peg} — roughly indifferent between the two corner regimes.
- **Services** (sticky, central, low import exposure): Managed > Peg > Float. The persistence channel (previous slide) hurts it more than direct FX exposure does, so it would rather fix the exchange rate than float it.

Key point: the sector that is most exposed to FX shocks directly (Resource) and the sector with the most at stake in the regime choice overall (Services) are not the same sector, and they even rank the two corner regimes in *opposite* order — a policymaker averaging across sectors could easily miss this.

Sensitivity: The Optimal Degree of FX Stabilization

Sensitivity sweep: re-solved the managed-float regime for $\phi_s \in [0, 2]$ ($\phi_s = 0$ is the pure float limit). ϕ_s does not affect the steady state, so this isolates the policy trade-off cleanly.



Reading:

- Going from pure float ($\phi_s = 0$) to just $\phi_s = 0.10$ more than halves the welfare loss ($10.48 \to 4.60$).
- The minimum is flat and wide: $\phi_s \in [0.2, 0.3]$ is statistically indistinguishable, and the calibrated value (0.30) is very close to optimal.
- Beyond $\phi_s \approx 0.3$, cost rises again — too much FX stabilization crowds out output-gap stabilization from the same single interest-rate instrument.

Bottom line: a little FX stabilization goes a long way; a lot is counterproductive. Neither corner regime (float or peg) is optimal — a modest, positive ϕ_s is.

Summary: Key Messages

1. **Free float is not the optimal regime, even though it directly targets the DC index.** Managed float has less than half the total welfare loss of pure float (4.19 vs. 10.48, $\times 10^4$) and beats the hard peg too (8.67). This is the model's own solution, not an assumption.
2. **Why:** floating turns the exchange rate into a slow, persistent state variable (via UIP and the near-unit-root NFA process) that feeds continuously back into every sector's marginal cost through Ω_i^F. This transforms even purely *domestic* TFP shocks into long, drawn-out inflation, whereas a peg resolves the same shock via a fast one-time price adjustment. **Persistence, not on-impact volatility, is what drives the welfare cost.**
3. **Every sector prefers managed float first**, but ranks the two corner regimes oppositely: Resource (import-exposed, flexible, peripheral) ranks float over peg; Services (import-light, sticky, central) ranks peg over float. The sector most exposed to FX shocks and the sector with the most at stake in the regime choice are different sectors.
4. **The optimal amount of FX stabilization is small but strictly positive**: welfare is minimized around $\phi_s \approx 0.2$–0.3, with more than half the gain captured by just $\phi_s = 0.10$; too much FX stabilization crowds out output-gap stabilization from the same instrument.
5. These are first-order (order-1) welfare comparisons (order-2 pruning fails here due to the price-level/ER unit root implied by an inflation-only Taylor rule) — rankings should be read as indicative of the mechanism, not certainty-equivalent-exact.

Next Steps

Open questions going forward:

- Sensitivity to calibration (alternative import shares Ω^F, Calvo probabilities δ_i): is managed-float dominance robust?
- A price-level-anchored policy rule, to obtain a fully second-order-accurate welfare ranking.
- Formalizing the exchange-rate persistence mechanism analytically.